Power Series, Explained Simply

1 What is a power series?

Some functions, like $\cos x$ and e^x, are hard to compute by hand. Polynomials, like $1 + x + x^2$, are easy: you only need to add and multiply.

A **power series** is like a polynomial that never stops. It looks like this:

$$a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

The a's are just numbers. Each term uses a higher power of x.

The big idea

Many complicated functions can be rewritten as an infinite sum of powers of x. If you stop after a few terms, you get a polynomial that can give a good approximation to the function.

A first example. If $|x| < 1$, then

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

Try $x = \frac{1}{2}$. The left side is $\frac{1}{1-1/2} = 2$. The right side is

$$1 + 0.5 + 0.25 + 0.125 + \cdots$$

which keeps getting closer to 2.

But try $x = 3$. The left side is fine ($\frac{1}{1-3} = -0.5$), but the right side is

$$1 + 3 + 9 + 27 + \cdots$$

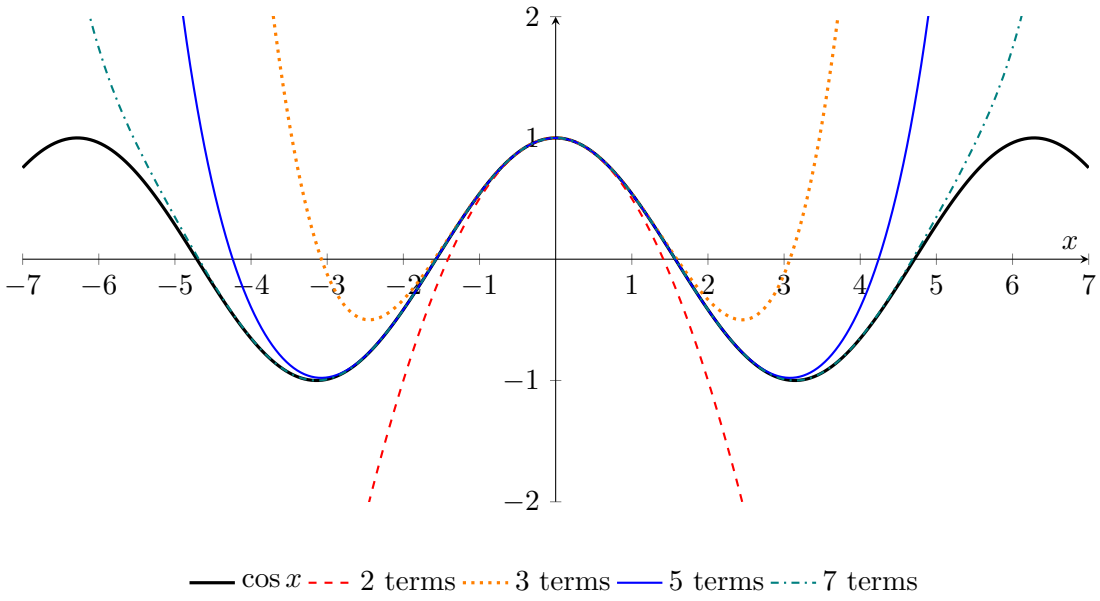
which just explodes. So the series only works for some values of x. Figuring out which ones is the second half of these notes.

2 The picture: building $\cos x$

Let's approximate $\cos x$ near $x = 0$ with simple polynomials, one step at a time.

- **Step 1:** 1. This is a flat line. It gets the starting height right ($\cos 0 = 1$) and nothing else.

- **Step 2:** $1 - \dfrac{x^2}{2}$. This is an upside-down parabola. It matches the top of the cosine hump.
- **Step 3:** $1 - \dfrac{x^2}{2} + \dfrac{x^4}{24}$. This bends back up on the sides, like the real cosine does.
- Keep adding terms. Each extra term improves the approximation near $x = 0$.



What to notice in the graph:

- Every curve sticks close to $\cos x$ near the middle.
- Adding more terms generally makes the approximation accurate over a wider range.
- Far away from zero, a finite polynomial approximation can become inaccurate. However, the complete infinite Taylor series for $\cos x$ actually converges to $\cos x$ for every real value of x.

Try it with numbers. The real value of $\cos 2$ is about -0.41615.

Terms used	Result at $x = 2$
1 term	1
2 terms	-1
3 terms	-0.333
4 terms	-0.422
5 terms	-0.4159
6 terms	-0.41615

After just six terms we are already very close to the actual answer!

3 Where do the numbers come from?

How did we know to use 1, then $-\frac{1}{2}$, then $\frac{1}{24}$? There is a recipe, and it comes from matching the function at one point.

The idea: build a polynomial that has the *same height, same slope, same curve, and so on* as your function at the center point. Each extra term matches one more of these.

The Taylor series recipe

For a function f centered at $x = 0$, the coefficient of x^n is

$$\frac{f^{(n)}(0)}{n!}$$

In words: take the nth derivative, plug in 0, and divide by $n!$ (that is, $n \times (n-1) \times \cdots \times 1$). The resulting Taylor series is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

whenever the series converges to the function.

More generally, if we center the series at $x = a$ instead of 0, we use

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

Again, this equality holds when the Taylor series represents the function.

Example with $\cos x$. Take derivatives and plug in 0:

$$\cos 0 = 1, \quad -\sin 0 = 0, \quad -\cos 0 = -1, \quad \sin 0 = 0,$$

$$\cos 0 = 1, \quad \ldots$$

The values go $1, 0, -1, 0, 1, 0, -1, \ldots$ and repeat. Divide each by the matching factorial and you get

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots$$

The zeros are why only even powers show up.

Series worth knowing:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

Shortcuts. You usually don't need to start from scratch. Within the radius of convergence, you can:

- **Swap x for something else.** Replace x with $-x^2$ in $\frac{1}{1-x}$ to get

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \cdots$$

- **Differentiate** each term separately.
- **Integrate** each term separately.

4 Where does it work? Radius and interval of convergence

Remember the example where the series for $\frac{1}{1-x}$ worked for $x = \frac{1}{2}$ but exploded for $x = 3$? Every power series has a region around its center where it converges, and we want to find it.

Key facts

- The region is centered at the middle of the series (usually 0).
- Its half-width is called the **radius of convergence**, written R.
- Inside the radius, the series converges absolutely.
- Outside the radius, the series diverges.
- The **interval of convergence** tells us exactly which real values of x make the series converge.
- The two endpoints are special and need to be checked separately.

There are three possibilities: the series works nowhere except at its center ($R = 0$), it works everywhere ($R = \infty$), or it works within some finite radius around its center.

How to find R: the ratio test

The test asks: *how quickly does each term shrink compared to the one before it?*

Compare each term to the previous one by dividing, and see what happens as n gets huge.

The ratio test

For a series $\sum u_n$, calculate

$$L = \lim_{n\to\infty} \left| \frac{u_{n+1}}{u_n} \right|$$

Then:

$$\begin{cases} L < 1 & \text{The series converges absolutely.} \\ L > 1 & \text{The series diverges.} \\ L = 1 & \text{The test is inconclusive.} \end{cases}$$

Remember: when $L = 1$, we cannot conclude anything from the ratio test alone.

Recipe for finding the radius

1. Write the ratio $\left|\dfrac{\text{next term}}{\text{current term}}\right|$ and simplify.
2. Find its limit as $n \to \infty$.
3. Set the limit less than 1 and solve for x. This gives the region of absolute convergence.
4. Check the two endpoints separately using other convergence tests where necessary.

Example 1: e^x. The terms are $\dfrac{x^n}{n!}$. The ratio is

$$\left|\frac{x^{n+1}/(n+1)!}{x^n/n!}\right| = \frac{|x|}{n+1}$$

As n gets huge, this goes to 0, which is less than 1 no matter what x is.

So the series works for every real x, and

$$R = \infty.$$

The same is true for $\cos x$ and $\sin x$.

Example 2: $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n}$.

The ratio is

$$\left|\frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n}\right| = |x|\frac{n}{n+1}$$

As $n \to \infty$, this approaches $|x|$.

We need

$$|x| < 1$$

so $R = 1$ and the initial region is $-1 < x < 1$.

Now check the endpoints:

- At $x = 1$, the series becomes
$$1 + \frac{1}{2} + \frac{1}{3} + \cdots$$
This is the harmonic series, which diverges.
- At $x = -1$, the series becomes
$$-1 + \frac{1}{2} - \frac{1}{3} + \cdots$$
This is an alternating series whose terms decrease to zero, so it converges by the alternating series test.

Therefore, the interval of convergence is

$$\boxed{[-1, 1)}$$

The square bracket means -1 is included, and the round bracket means 1 is left out.

Example 3: not centered at 0.

Consider

$$\sum_{n=1}^{\infty} \frac{(x-2)^n}{n\,3^n}.$$

The center is 2. The ratio is

$$\left| \frac{(x-2)^{n+1}}{(n+1)3^{n+1}} \cdot \frac{n3^n}{(x-2)^n} \right| = \frac{|x-2|}{3} \frac{n}{n+1}$$

As $n \to \infty$, this approaches

$$\frac{|x-2|}{3}.$$

We need

$$\frac{|x-2|}{3} < 1$$

which gives

$$|x-2| < 3.$$

Therefore, $R = 3$ and the initial region is

$$-1 < x < 5.$$

Checking the endpoints:

- At $x = 5$, we get $\sum_{n=1}^{\infty} \frac{1}{n}$, which diverges.
- At $x = -1$, we get $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$, which converges by the alternating series test.

Therefore, the interval of convergence is

$$\boxed{[-1, 5)}.$$

Example 4: both endpoints can converge.

Consider

$$\sum_{n=1}^{\infty} \frac{x^n}{n^2}.$$

Using the ratio test:

$$L = \lim_{n \to \infty} \left| \frac{x^{n+1}}{(n+1)^2} \cdot \frac{n^2}{x^n} \right|$$

$$= |x| \lim_{n\to\infty} \frac{n^2}{(n+1)^2} = |x|.$$

We need $|x| < 1$, so $R = 1$.

Now check the endpoints:

- At $x = 1$:
$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$
This converges because it is a p-series with $p = 2 > 1$.
- At $x = -1$:
$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$
This also converges absolutely because
$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$
converges.

Therefore, the interval of convergence is
$$\boxed{[-1, 1]}.$$

Notice how this differs from Example 2: here both endpoints are included!

5 Quick summary

- A power series is an infinite sum of powers of x.
- Stopping after a few terms gives a polynomial approximation.
- The coefficients of a Taylor series come from derivatives at the center:
$$\frac{f^{(n)}(a)}{n!}$$
- The radius of convergence tells us how far from the center the series converges absolutely.
- The ratio test helps us find the radius of convergence.
- If the ratio test gives $L = 1$, it is inconclusive.
- Always check the two endpoints separately.

Practice

Find the radius and interval of convergence for each series. Remember to show your ratio test work and check the endpoints!

(1) $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{2^n}$

(2) $\displaystyle\sum_{n=1}^{\infty} \frac{(x+1)^n}{\sqrt{n}}$

(3) $\displaystyle\sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$

(4) $\displaystyle\sum_{n=1}^{\infty} \frac{(x-1)^n}{n\,2^n}$

Answers:

(1) $R = 2$, $(-2, 2)$.
(2) $R = 1$, $[-2, 0)$.
(3) $R = \infty$, $(-\infty, \infty)$.
(4) $R = 2$, $[-1, 3)$.